Reflection on EE517

During the Spring 2026 semester, I took the course EE517 - AI Accelerated Computing, taught by Dr. Fam. The course description did an excellent job of luring me in, as it mentioned the topics of Matrix multiplication, and fast algebra. Entering the course, my objective was to gain knowledge on the mathematization of computing for AI-related applications. Dr. Fam's teaching and teaching and papers have exceeded my expectations, by providing insight into the history of this field of algebra. During the lectures, we studied papers including Winograd's "A New Algorithm for Inner Product[1], Karatsoba's "Multiplication of Many-Digital Numbers by Automatic Computers" [2] and Dr. Fam's "Optimal Partitioning and Redundancy Removal in Computing Partial Sums"[3]. Dr. Fam did an excellent job of exposing not only the content of these papers, but also how they were created, and how they may be applied to our current state of affairs in today's world of AI computing.

One technical aspect of this course I shall now focus upon is the aforementioned 'Optimal Partitioning and Redundancy Removal in computing partial sums.' by Dr. Fam. This paper has been a central topic of our lectures in the course. The summary of this paper is that it proposes two algorithms for computing large partial sums in parallel, and suggests hardware architectures for their implementation. The algorithms successfully reduce the number of computations from D to O(D/log(D)). I will now focus on mathematically conveying the first algorithm, which reduces computations by means of output partitioning.

For this output partitioning algorithm, the output Y = BX + U, where B is a Matrix and X is a vector. The matrix B is composed of 1s and 0s so that the outputs y_1 through y_n assume forms where they consist of sums of x_1 through x_n. After formatting the output, we separate the first output, and rewrite the matrix B. The next step is to add the identical columns of B and list them as s_1 through s_n. These s values are multiplied by a new smaller B matrix, which has a number of columns equal to the value of n from s_n. B was reduced by removing the redundant identical columns. This completes one iteration of the algorithm. These iterations are repeated until the final iteration results in 1 output that contains only 1 addition. An overview of the process is shown in Figure 1. After the output partitioning algorithm, the other algorithm which reduces computations by means of input partitioning is presented. Lastly, hardware architecture is suggested for implementing these algorithms, including structured adder trees.

Computing Partial Sums

1) Output partitioning $B = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ \vdots \\ x_7 \end{bmatrix}$, $U = \begin{bmatrix} u_1 \\ \vdots \\ u_3 \end{bmatrix}$

★ note, x contains set of values the output is composed of ★

$y_1 = u_1 + x_4 + x_5 + x_6 + x_7$

$y_2 = u_2 + x_2 + x_3 + x_6 + x_7$

$y_3 = u_3 + x_1 + x_3 + x_5$

Step 1

$y_1 = (((u_1 + x_4) + x_5) + x_6) + x_7$ (4 adds)

new B: $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$

Step 2

remove x_4 (0-column), add inputs with identical columns

$s_1 = x_1 + x_5$

$s_2 = x_2 + x_6$

$s_3 = x_3 + x_7$

★ The 3 possible forms ★

results in $\begin{bmatrix} y_2 \\ y_3 \end{bmatrix} = B^{(1)} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} + \begin{bmatrix} u_2 \\ u_3 \end{bmatrix}$, $B^{(1)} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

Iteration 2

$y_2 = u_2 + s_2 + s_3$

$t_{23} = u_2 + s_2 \qquad y_2 = t_{23} + s_3$ (2 adds)

$B^{(1)} \longrightarrow [1\ 0\ 1]$

$t_1 = s_1 + s_3 \qquad r = 1,\ n = 1$

$y[3] = 1\,t_1 + u_3 \Rightarrow y_3 = u_3 + t_1$

Iteration 3

$y_3 = u_3 + t_1 \qquad B = [1]$ (1 add)

$4 + 3 + 1 = 8$ adds

Figure 1.

The output partitioning algorithm is one example of the many thought-provoking and important topics during the lectures throughout the semester. This course is one of the best I've ever taken because I have learned not only about these topics from a mathematical perspective, but how to put myself in the same frame of mind as those who first researched and discovered these mathematical relations. I am thankful that Dr. Fam has emphasized that as a student I need to think analytically and derive solutions on my own so that I may completely understand them. While I may not be able to speak with Karatsoba or Winograd as I can with Dr. Fam, I can still study hard to have the same moment of discovery that they did.

1. S. Winograd, "A new algorithm for inner product," *IEEE Transactions on Computers*, vol. 17, no. 7, pp. 693–694, 1968
2. A. A. Karatsuba and Y. P. Ofman, "Multiplication of many-digital numbers by automatic computers," *Proceedings of the USSR Academy of Sciences*, vol. 145, pp. 293–294, 1962.
3. A. T. Fam, "Optimal partitioning and redundancy removal in computing partial sums," *IEEE Transactions on Computers*, vol. C-36, no. 10, pp. 1137–1143, Oct. 1987.